Discrete Structures

Problem Set 3: Groups and Rings

Release Date: October 16th, 2026
Deadline: October 31st, 2026

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1 Instructions

- The marks for each question (or sub-question) are specified alongside it.
- If you use variables that are not defined in the question, state them clearly, and the domain they belong to as well (if applicable).
- Answer questions precisely, maintaining mathematical rigour as much as possible.
- Both digital (written, not typed) and handwritten submissions will be accepted.
- The deadline for this set is October 31st, 11:59 p.m.
- Submissions to be done on Moodle with the format `{rollnumber}_DS_A3` (for example `2025113019_DS_A3`).
- **Personalised question.** Question 2 depends on your roll number. Write the value of v at the top of your answer to Question 2; an answer to the wrong variant receives no credit.
- Plagiarism and AI use shall be penalised heavily.

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2 Groups

Question 1 **[16 marks]**

Let G be a non-empty set with an associative binary operation, written as juxtaposition.

(a) Suppose that for all $a, b \in G$ both equations $ax = b$ and $ya = b$ have solutions $x, y \in G$. Prove that G is a group. **[6]**

(b) Now let G be a group in which x^2 is the identity element for every $x \in G$. Prove that G is abelian. **[5]**

(c) Let G be any group. Prove that the map $\phi(x) = x^{-1}$ is a homomorphism from G to G if and only if G is abelian. **[5]**

Question 2 **[18 marks]**

(Personalised.) Let d be the last digit of your roll number and let $v = d \bmod 4$. Take n and H from the table below.

v	0	1	2	3
n	32	40	48	60
H	$\{1, 31\}$	$\{1, 9\}$	$\{1, 47\}$	$\{1, 49\}$

(a) Verify that H is a subgroup of $U(n)$, use Lagrange's theorem to find the number of cosets of H in $U(n)$, and list them all. **[4]**

(b) Explain why the cosets form a group $U(n)/H$ under $(aH)(bH) = abH$, and write out its Cayley table. **[6]**

(c) Exactly one of $\mathbb{Z}_8$ (under addition mod 8), $U(16)$ and $U(24)$ is isomorphic to $U(n)/H$. Say which, give an explicit isomorphism and verify it is one, and prove $U(n)/H$ is isomorphic to neither of the other two. **[8]**

Question 3 **[16 marks]**

Throughout, $U(n)$ is under multiplication modulo n and $\mathbb{Z}_8$ is under addition modulo 8.

(a) Let $\phi : G \to G'$ be an isomorphism. Prove that for every $x \in G$ of finite order, x and $\phi(x)$ have the same order. **[4]**

(b) Show that $U(16)$ and $U(20)$ are isomorphic by giving an explicit isomorphism (a table or a formula) and verifying it. **[7]**

(c) Prove that neither $U(16)$ nor $U(20)$ is isomorphic to $\mathbb{Z}_8$ or to $U(24)$. **[5]**

3 Rings

Question 4 **[18 marks]**

For a prime p, let $\mathbb{Z}_p[i] = \{a + bi : a, b \in \mathbb{Z}_p\}$ with $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$, all modulo p. You may assume it is a commutative ring with unity.

(a) Let R be a finite commutative ring with unity. Prove that every non-zero element of R is either a unit or a zero divisor (and not both). **[5]**

(b) For an odd prime p, prove that $\mathbb{Z}_p[i]$ is a field if and only if -1 is not a square in $\mathbb{Z}_p$. **[7]**

(c) For $p = 5$, list all the zero divisors of $\mathbb{Z}_p[i]$ and find the inverse of $1 + i$. **[6]**

Question 5 **[16 marks]**

Let $F = \mathbb{Z}_2[x]/\langle x^2 + x + 1\rangle$.

(a) Show that F has exactly four elements, and write out its addition and multiplication tables. **[6]**

(b) Prove that F is a field and that its non-zero elements form a cyclic group under multiplication. **[5]**

(c) Show that F has characteristic 2, and deduce that $\varphi(a) = a^2$ is a ring isomorphism from F to itself. **[5]**

Question 6 **[16 marks]**

Let $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$ be the Gaussian integers and $N(a + bi) = a^2 + b^2$. You may use that N is multiplicative and that $\mathbb{Z}[i]$ is an integral domain.

(a) Prove that for all $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$ there exist $q, r \in \mathbb{Z}[i]$ with $\alpha = q\beta + r$ and $N(r) < N(\beta)$. **[6]**

(b) Deduce that every ideal of $\mathbb{Z}[i]$ is principal. **[5]**

(c) Show that $\langle 3 + i, 5\rangle = \langle 2 - i\rangle$, where $\langle \alpha, \beta\rangle = \{x\alpha + y\beta : x, y \in \mathbb{Z}[i]\}$. **[5]**